Sensory Innervation of Ocular Tissues

Images used: 13.1, 13.4, 13.3, 10.18, 13.5, 13.6, 13.7

I. Sensory innervation: Trigeminal nerve (cranial nerve 5)

A. Ophthalmic division of the trigeminal nerve

1. Nasociliary nerve is formed by the combination of 5 branches.

a. Infratrochlear nerve
↳ under

(1) Innervation of
caruncle
canaliculi / lac. sac
med lids + conj.

b. Anterior ethmoid nerve

(1) Innervation skin of medial nose
ethmoid sinus

(2) Clinical correlate: herpes zoster

i. Hutchinson sign: lesion on tip of nose

c. Posterior ethmoid nerve

(1) Innervation of ethmoid + sphenoid sinus

d. Long ciliary nerves

(1) Innervation of ant. segment
ant sclera, iris, CB, cornea + TM
(AI shorts check)

e. Short ciliary nerves

(1) Innervation ant. segment
cornea + iris + ciliary body

(2) Pathway

i. Pass through the ciliary ganglion as the sensory root of the ciliary ganglion

2. Frontal nerve

    a. Supratrochlear nerve

        (1) Innervation: forehead
        medial upper eyelid

    b. Supraorbital nerve

        (1) Innervation: forehead
        central upper eyelid

3. Lacrimal nerve

    a. Innervation: lacrimal gland
    lateral upper eyelid
    temple area

4. The nasociliary, frontal, and lacrimal nerves join to form the ophthalmic division of the trigeminal nerve.

B. Maxillary division of the trigeminal nerve

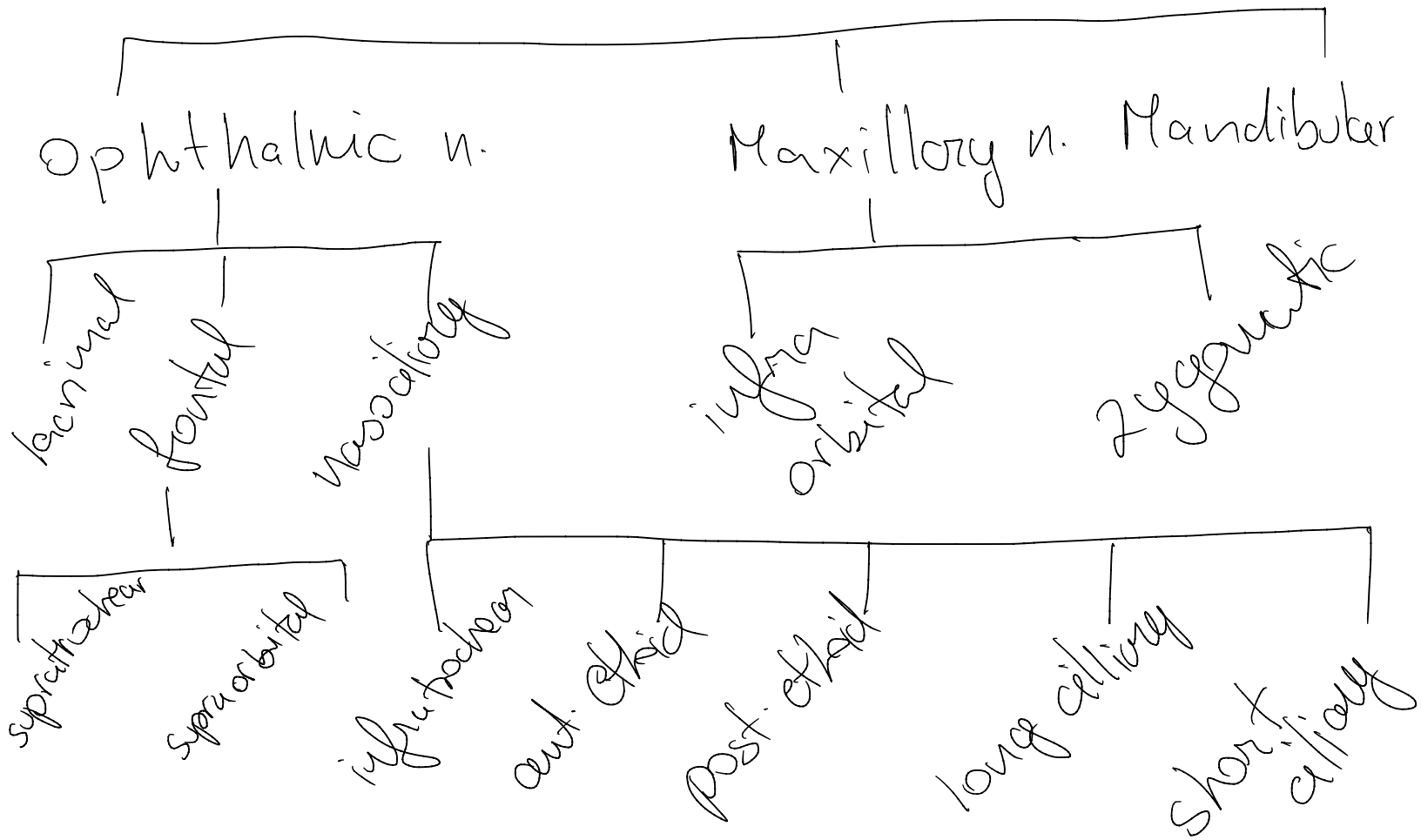
1. Infraorbital nerve

    a. Innervation lower eyelid
    upper cheek, upper lip + teeth

2. Zygomatic nerve

    a. Innervation: lateral lower eyelid
    lateral cheek
    lateral forehead

TRIGEMINAL NERVE



Autonomic Innervation to Ocular Tissues

Figures used: 14.2, 14.3

I. Sympathetic pathway to the internal ocular structures

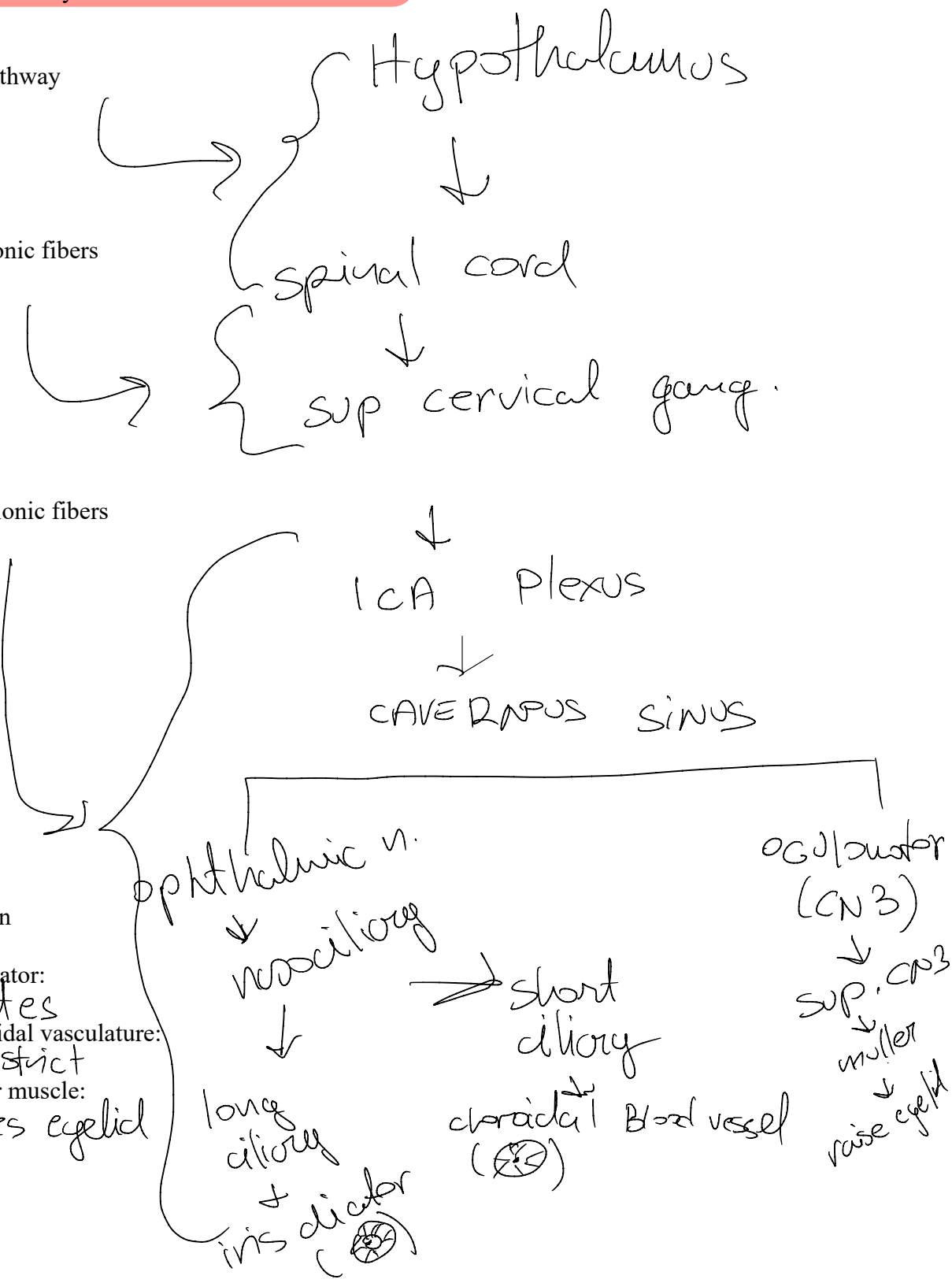
A. Central pathway

B. Preganglionic fibers

C. Postganglionic fibers

D. Innervation

1. Iris dilator: dilates
2. Choroidal vasculature: constrict
3. Muller muscle: raises eyelid

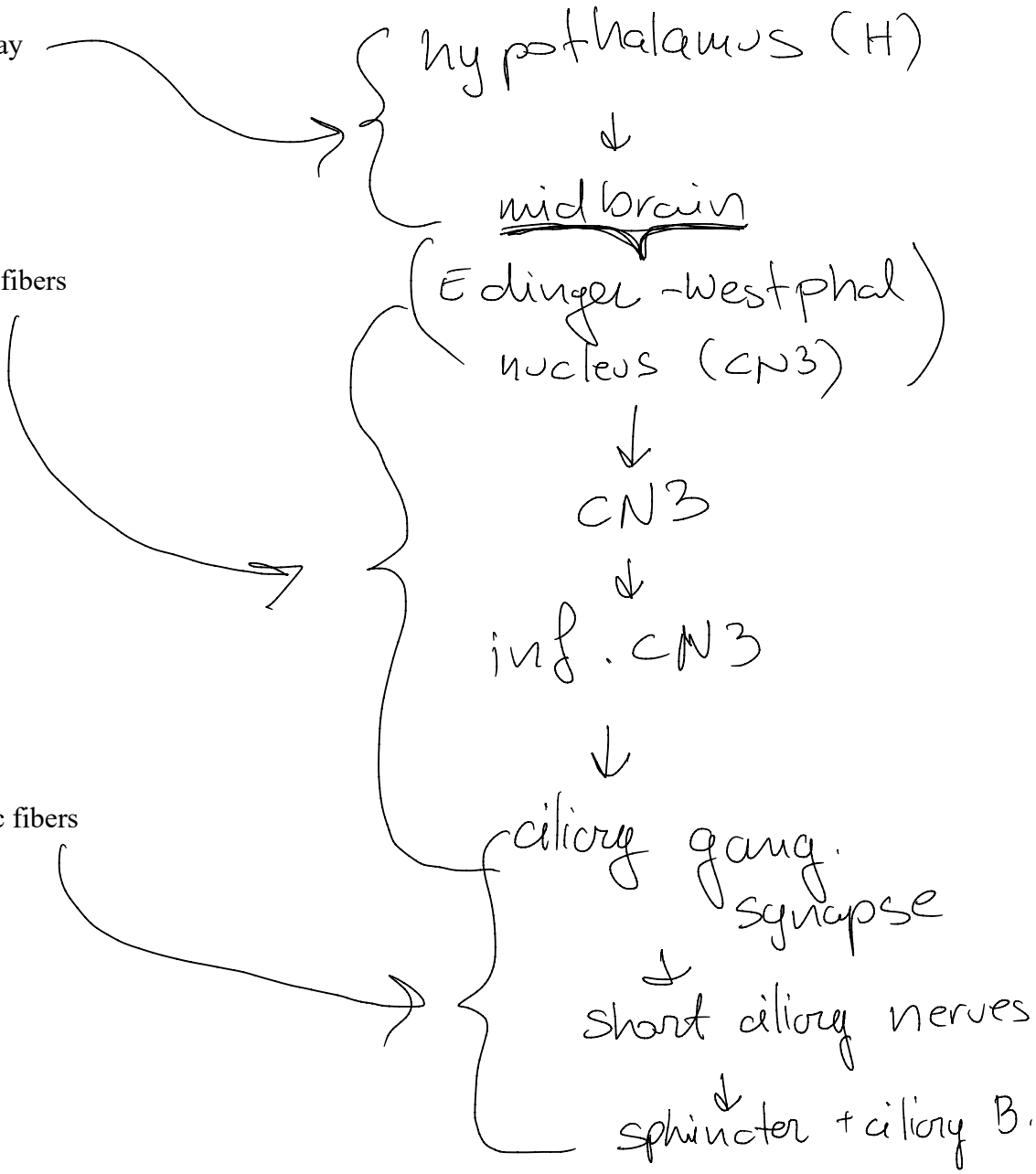


II. Parasympathetic pathway to the internal ocular structures

A. Central pathway

B. Preganglionic fibers

C. Postganglionic fibers



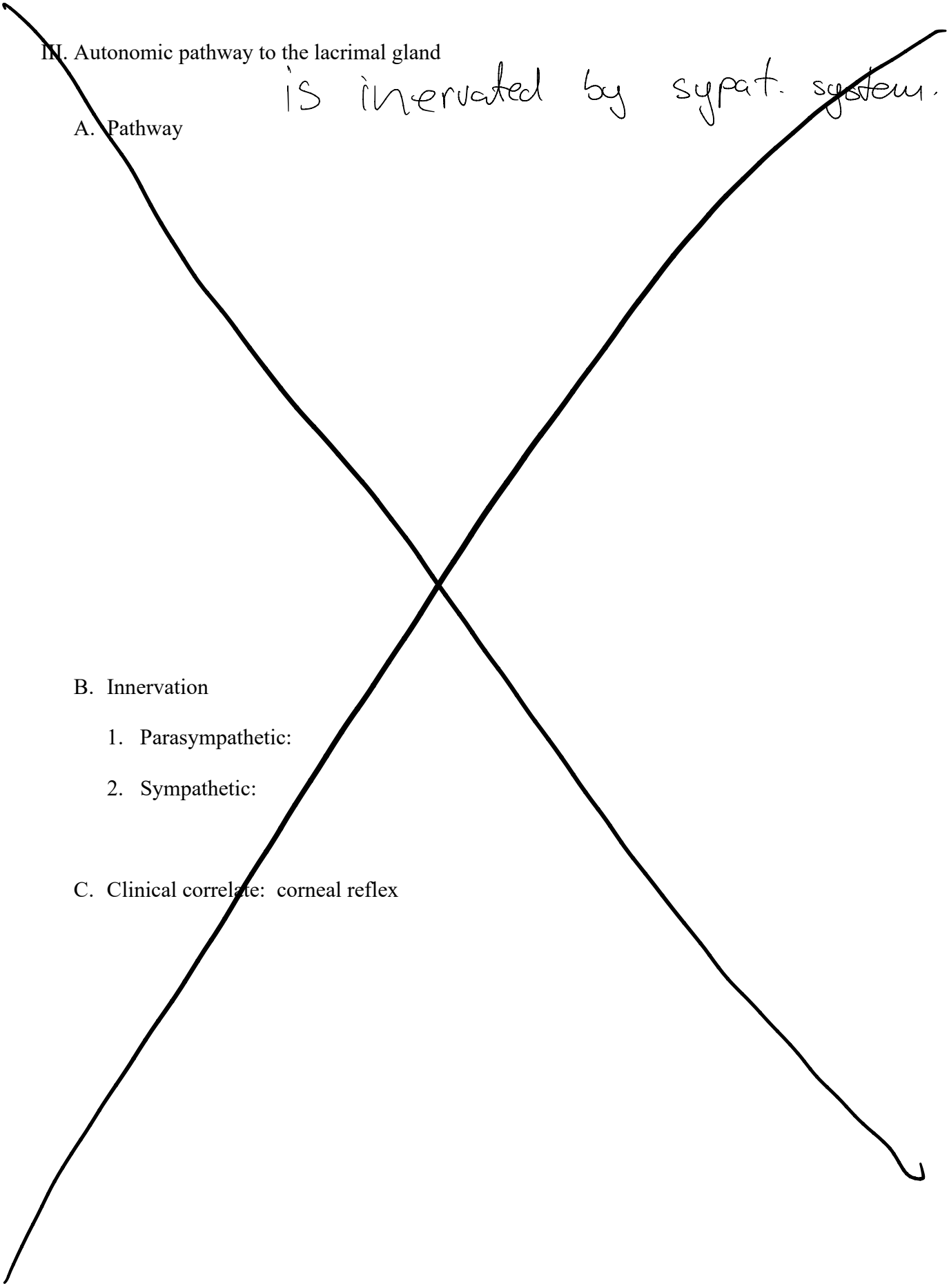
D. Innervation

1. Iris sphincter: miosis
2. Ciliary muscles: accommodation

III. Autonomic pathway to the lacrimal gland

is innervated by sypat. system.

A. Pathway



B. Innervation

1. Parasympathetic:
2. Sympathetic:

C. Clinical correlate: corneal reflex

Skeletal Motor Innervation to Ocular Tissues

Images used: 13.10, 13.7, 10.18, 13.11, 13.12, 13.13, 13.14, 13.16, 13.17, 13.18

(cranial nerve = CN) CN3, CN4, CN6
CN7 → orbo

I. Oculomotor nerve (cranial nerve 3)

A. Innervates Superior rectus, medium rectus, inferior oblique, inferior rectus, levator } they go to sphincter, CB to accomodation

B. Oculomotor nucleus: superior midbrain

C. Cranial nerve 3 pathway

1. Leaves the brainstem at the level of the midbrain.

2. Passes between some important arteries.
   superior cerebellar & post. arteries
3. Follows the posterior communicating A. } signs of a stroke? (aneurizma)

   a. Clinical correlate: CN3 is most frequently affected by posterior communication artery aneurysm but can be affected by any of the surrounding arteries.

4. Cavernous sinus if something happens here, eye stays straight and we don't feel anything

5. Superior orbital fissure Whole that allows all the nerve guys go through that hole

6. Orbit

    a. Superior branch: innervates the SR and levator muscle

Superior goes Muller through superior branch everything else goes through INFERIOR BRANCH

    b. Inferior branch

MR | IR | IO | Para Symp.

D. Clinical correlate: cranial nerve 3 palsy

    a. Ptosis

    b. The eye is positioned down and out due to the unopposed actions of the SO and LR

    c. They cannot adduct (MR)

    d. In the abducted position, the eye cannot move up or down (SR, IR)

    e. In the adducted position, the eye cannot move the eye up (IO)

    f. Pupil dilation

    g. Decreased accommodation

II. Trochlear nerve (cranial nerve 4)

A. Innervates: superior oblique

B. Nucleus: inferior midbrain

C. Cranial nerve 4 pathway

1. Leaves the brainstem from the posterior midbrain and then crosses to the opposite side.

2. Follows the superior cerebellar and posterior cerebral arteries around the brainstem

3. Cavernous sinus

4. Superior orbital fissure

5. Orbit

D. Clinical correlate: cranial nerve 4 palsy

1. Eye deviation in primary position: hyperdeviation

2. Restrictions: would not be able to depress in adduction

III. Abducens nerve (cranial nerve 6)

A. Innervates: lateral rectus (LR)

B. Nucleus: pons

C. Cranial nerve 6 pathway

1. Exits the groove between the pons and medulla

2. Runs along the occipital bone and posterior slope of the petrous portion of the temporal bone

   a. Makes a sharp bend over the petrous ridge

   b. Clinical correlate: Susceptible to trauma and increased intracranial pressure.

3. Cavernous sinus

4. Superior orbital fissure

5. Orbit

D. Clinical correlate: CN 6 palsy

1. Eye deviation in primary position: eso

2. Restrictions: abduction defect